

Scrap has patterns.

What if every production run could be scored before the press starts?

Decision Tree · Multiclass Risk · Process Rules



2,317

Production Runs

8

Process Features

72.0%

Accuracy

3

Risk Classes

Scrap in a stamping line is not random — it is the compound result of identifiable conditions converging at the wrong time.



Material Cost

Scrapped stampings waste raw material, tooling wear, and cycle time with zero output



Line Disruption

High scrap rates trigger stops, rework loops, and downstream scheduling failures



Invisible Causes

No single variable causes scrap — it's the combination of operator, speed, material, and setup

53.5%

of runs with a critical supplier lot are classified High Risk — vs 19.4% for standard lots. The material matters as much as the operator.

Old Thinking

Scrap is discovered after the fact:

"It happened / we scrapped it."

Check the bin at end of shift.
Log the defect. File the report.
No prevention. No pre-run decision.
Just counting what was already lost.



New Thinking

Risk is scored before the run begins:

"How likely is High Risk — right now?"

Score the run before the press starts.
Flag High Risk → adjust or intervene.
Complete the checklist, reduce speed.
Turn a prediction into a process decision.

Eight process variables across four operational layers — each one a lever that process engineers already control:

PRESS	Press speed (spm), Raw material hardness (HRB) <i>Machine parameters — the physical conditions of each production run</i>
OPERATOR	Operator experience (years), Shift (Day/Night/Early Morning) <i>Human factors — experience level and time-of-day operational context</i>
MATERIAL	Critical supplier lot flag, Ambient temperature (°C) <i>Input quality and environmental conditions at run start</i>
PROCESS	Recent model change flag, Setup checklist complete flag <i>Process governance — whether pre-run protocols were followed</i>
Low: 19.9% Medium: 50.4% High: 29.7%	

Every variable here already exists in the production system — no new sensors, no new data collection.

A model that prints rules an engineer can tape to the press is more valuable than one that only runs in a Jupyter notebook.

No Scaling Required

Trees split on thresholds, not distances. StandardScaler has zero effect on a Decision Tree — simpler preprocessing, fewer failure points in production.

Human-Readable Rules

`export_text()` outputs if-then-else rules that can be posted as a process control document. The model IS the SOP — not a proxy for one.

Interaction Capture

Velocity + inexperienced operator + incomplete checklist = High Risk. Trees capture this multiplicative logic naturally without feature engineering.

Controlled Complexity

`max_depth=5`, `min_samples_leaf=50` prevents overfitting. Each leaf represents ≥ 50 production runs — not a single outlier. Regularization by design.

The best model for manufacturing is the one the process engineer will actually use to make a decision before the shift starts.

72.0%

Test Accuracy

71.8%

F1 Macro

23.7%

Top Feature Imp.

5

Max Tree Depth

Gini Feature Importance

operator_experience_yrs	23.2%
setup_checklist_complete	23.7%
critical_supplier_lot	19.2%
press_speed_spm	17.3%
recent_model_change	16.6%
shift / hardness / temp	0.0%

Confusion Matrix (696 test runs)

	P:High	P:Low	P:Med
A:High	154	0	53
A:Low	3	99	36
A:Med	59	44	248

Most errors are directionally conservative: Medium runs scored as High (59) and Low runs scored as Medium (36). Over-flagging risk is the safer failure mode.

F1 Macro treats all three risk levels equally — the right metric when missing any class has operational consequences.

Operator Exp. 23.2%

→ Structure operator pairing: inexperienced operators (< 2 yrs) should not run high-speed setups on critical supplier lots without senior oversight.

Setup Checklist 23.7%

→ Checklist completion is non-negotiable — incomplete runs show 42.2% High Risk vs 16.8%. Make it a physical gate, not a checkbox after the fact.

Supplier Lot 19.2%

→ Critical lots trigger a mandatory speed reduction and experienced-operator assignment. Route flagged batches to controlled cells only.

Press Speed 17.3%

→ High speed + inexperienced operator is the dominant interaction. Set conditional speed limits by operator tier — not a single universal maximum.

Shift / Hardness / Temp 0%

→ These features carry zero Gini importance in this tree. Night shift effect is already captured by operator experience and checklist completion rates.

What-If Simulator — Pre-Run Risk Scorecard

08

Same press. Same shift. Three different setups. The third one shows exactly which intervention flips the outcome.

Run A — Low Risk	Run B — High Risk	Run C — Intervention
Speed 30 spm	Speed 55 spm	Speed 38 spm
Hardness 80 HRB	Hardness 92 HRB	Hardness 92 HRB
Operator 10 yrs	Operator 0.5 yrs	Operator 0.5 yrs
Supplier Standard	Supplier Critical	Supplier Critical
Checklist ✓ Complete	Checklist X Incomplete	Checklist ✓ Complete ↑
Shift Day	Shift Night	Shift Night
 LOW — 89.7%	 HIGH — 91.6%	 MEDIUM — 55.6%

Run C insight: speed reduction (55→38 spm) + completed checklist shifts outcome from High (91.6%) to Medium (55.6%). Two interventions, same material, same operator.

The simulator is a go/no-go tool — not for logging what happened, but for deciding what to change before it does.

This project is not about predicting scrap.

It is about making the risk visible before the press starts.

Decision Trees don't find hidden patterns —
they formalize what experienced engineers already know, and make it auditable.

Feel free to download the full project: dataset, notebook, and simulator.

Change the depth. Adjust the leaf size. Challenge the tree's rules.

Because process quality has no finish line.

